

ARXIV 2405.04434 · DEEPSEEK-AI

DeepSeek-V2: A Strong, Economical, and Efficient MoE Language Model

236B

Total Parameters

21B

Activated per Token

128K

Context Length

8.1T

Pretraining Tokens

Two core innovations: Multi-Head Latent Attention (MLA) · DeepSeekMoE



CHALLENGE

The LLM Scaling Dilemma: Performance vs. Cost & Speed

Dense Model Costs

- Parameters grow proportionally with capability
- All weights activated per forward pass
- Training costs scale prohibitively with size
- KV cache grows linearly with heads × sequence length

MoE Promise

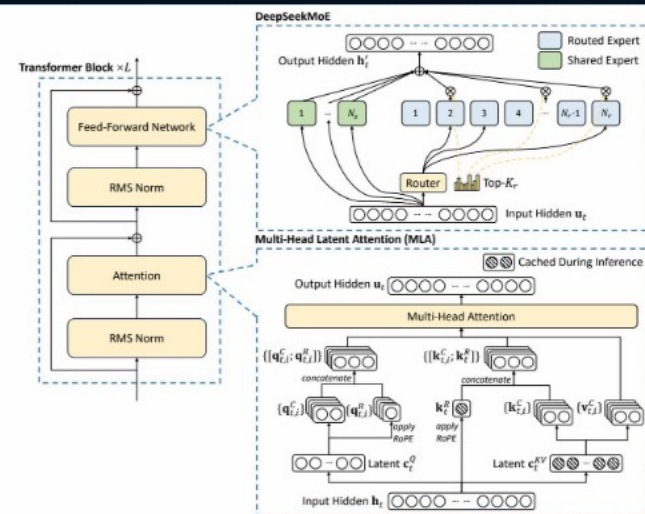
- Sparse activation: only a fraction of experts fire per token
- Scale total params without scaling compute proportionally
- Still bottlenecked by attention KV cache at long contexts
- GQA/MQA reduce cache but sacrifice model quality

DEEPSEEK-V2 Combines sparse MoE (DeepSeekMoE) with a novel KV-efficient attention (MLA) to achieve top-tier performance at a fraction of inference and training cost.

ARCHITECTURE

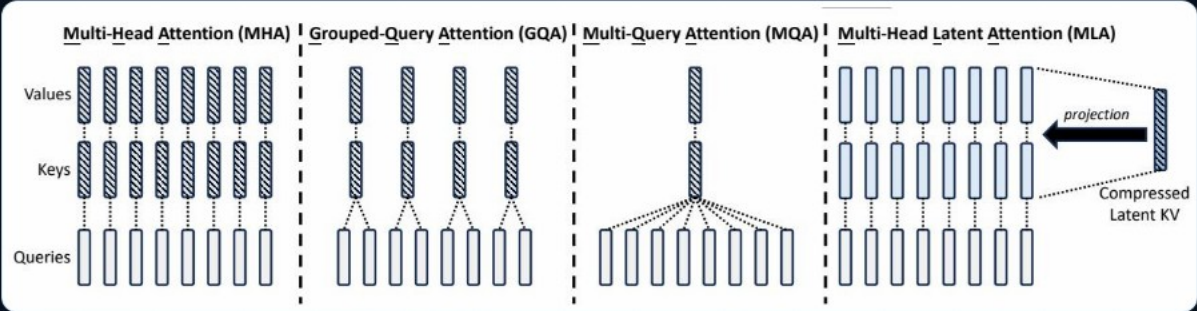
DeepSeek-V2: MLA + DeepSeekMoE in Each Transformer Block

- **MLA replaces Multi-Head Attention** — low-rank KV compression reduces cache by 93.3% while outperforming standard MHA
- **DeepSeekMoE replaces dense FFN** — shared experts (always active) + routed experts (sparse top-K activation)
- **Device-limited routing** constrains each token to experts on at most M devices, minimizing cross-node communication
- **236B total / 21B activated** — only 8.9% of parameters fire per token
- **128K context window** supported via decoupled RoPE positional embeddings in MLA



INNOVATION 1

Multi-Head Latent Attention: 93.3% KV Cache Compression



Standard MHA Bottleneck

Caches full K and V vectors per head per token — memory scales with $n_heads \times seq_len \times d_head$

GQA / MQA Trade-off

Share keys/values across heads to reduce cache, but sacrifice representational capacity and model quality

MLA: Low-Rank KV Joint Compression

Compress K and V into a single compact latent vector $c_t \cdot KV$. Cache only the latent; reconstruct K, V, and decoupled RoPE keys on-the-fly via learned projections

Result: Strictly Better Than MHA

93.3% KV cache reduction *with superior performance* — not a trade-off but a Pareto improvement over GQA/MQA

INNOVATION 2

DeepSeekMoE: Fine-Grained Experts with Shared Expert Isolation

NS SHARED EXPERTS

Always Activated

Capture **common knowledge** across all tokens. Always active regardless of routing decisions — stable representations for universal features

DEVICE-LIMITED ROUTING

Communication Efficiency

Each token's experts are constrained to at most **M devices** — drastically reduces cross-node all-to-all communication overhead

NR ROUTED EXPERTS

Sparse Top-K Activation

Sparsely activated via a **top-K_r router**. Fine-grained segmentation increases specialization potential per expert

AUXILIARY LOSSES

Load Balance & Token Drop

Auxiliary losses enforce **balanced expert utilization**. Token-dropping strategy during training prevents overflow without affecting inference

FFN output: **shared experts (Ns) + top-K_r routed experts from Nr total** · Device constraint: max **M devices** per token



EFFICIENCY

42.5% Training Cost Reduction & 5.76× Generation Throughput

All metrics relative to DeepSeek 67B dense predecessor

DeepSeek 67B → DeepSeek-V2

Metric	DeepSeek 67B (Dense)	DeepSeek-V2	Improvement
Training Cost	Baseline	-42.5%	↓ 42.5% reduction
KV Cache Size	Baseline	-93.3%	↓ 93.3% reduction
Max Generation Throughput	Baseline	5.76× higher	↑ 5.76× faster

Architecture Driver

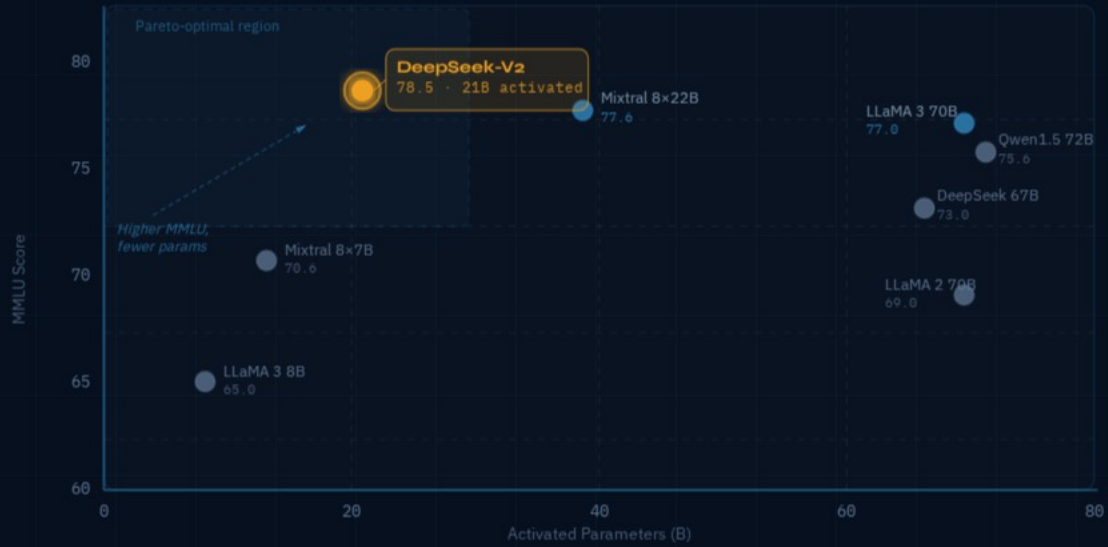
MLA eliminates redundant KV storage; DeepSeekMoE activates only 21B of 236B params per token

Inference Impact

5.76× throughput improvement enables practical large-scale deployment at significantly reduced serving cost

BENCHMARK

DeepSeek-V2 Tops MMLU at Only 21B Activated Parameters



ALIGNMENT

DeepSeek-V2 Chat (RL): Rivals Closed-Source on Chinese Benchmarks

SFT	Supervised Fine-Tuning 1.5 million conversational sessions spanning instruction following, coding, math, and reasoning domains
RL	Group Relative Policy Optimization (GRPO) Reinforcement learning without a separate critic model — groups of completions scored relative to each other, reducing compute vs. standard PPO

DeepSeek-V2 Chat (RL) outperforms all open-source models and most closed-source models on AlignBench (Chinese), demonstrating strong multilingual alignment.

AlpacaEval 2.0 Length-controlled win rate	38.9%
MT-Bench Overall score (out of 10)	8.97
AlignBench Overall — Chinese evaluation	7.91
🏆 Best open-source Chinese alignment — and competitive with GPT-4-class models on open-ended English evaluation	



SUMMARY

Key Takeaways: Architectural Innovation Unlocks Efficient Scaling



MLA: Superior KV Cache Solution

Low-rank joint compression reduces KV cache by **93.3%** while outperforming standard MHA — a strict Pareto improvement over GQA/MQA



Pareto-Optimal Performance

Only **21B activated params** match or exceed 70B-class dense models (LLaMA 3 70B, Qwen1.5 72B) on MMLU — highest efficiency frontier



DeepSeekMoE: Economical Training

Fine-grained experts + shared expert isolation enables 236B-param model at **42.5% lower training cost** than a 67B dense baseline



Multilingual Alignment Leadership

DeepSeek-V2 Chat (RL) achieves **MT-Bench 8.97** and best-in-class Chinese AlignBench score, competitive with closed-source systems

DeepSeek-V2 demonstrates that **architectural efficiency** and **top-tier performance** are not in tension — MLA + DeepSeekMoE unlock both simultaneously · [arXiv 2405.04434](#)

